

Rediscovering Physical Laws with Symbolic Regression

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1 What I did

I used symbolic regression (the PySR package) to see how well it can recover physical laws directly from data. Symbolic regression is different from a normal neural network: instead of a black box, it returns an actual equation, so you can read off the physics. I ran a series of experiments, starting with a clean test case (Kepler’s third law) and ending with real astronomical data where I did not know the answer ahead of time.

2 Kepler’s third law from simulated orbits

First I checked that the method works at all. I simulated orbits under Newton’s gravity ($F = -GM/r^2$) by integrating the equations of motion, and *measured* each orbit’s period from the trajectory — I never told the code the period formula. I then gave PySR the semi-major axis a and asked it to predict the period T . It recovered

$$T \propto a^{3/2}, \tag{1}$$

which is Kepler’s third law, to a precision of about 10^{-11} . So the relationship really emerges from Newton’s law; it was not assumed.

3 How much noise can it handle?

Real measurements are noisy, so I added increasing random noise to the periods and asked how often PySR still found the $a^{3/2}$ law. To make this fair I repeated each noise level over 10 random seeds and recorded the fraction of runs that recovered the law. The law was recovered reliably up to about 12% noise, became 50/50 around 20%, and was not recovered at all by 30%. Interestingly, when it failed it did not return a slightly-wrong exponent — it abandoned the power law entirely and returned a straight line. This is because PySR prefers simple equations, so once the noise is large enough that the (more complex) true law is no longer clearly better, it falls back to a simpler wrong one.

4 Two variables: adding the central mass

Next I let the central mass M vary as well, so the true law is $T = 2\pi a^{3/2}/\sqrt{GM}$. PySR correctly found that the period depends on *both* variables with the right powers:

$$T \propto a^{3/2} M^{-1/2}. \tag{2}$$

Recovering a positive 3/2 power on a and a negative 1/2 power on M at the same time is a real step up for PySR from fitting a single-variable curve.

5 A small deviation: does it notice modified gravity?

I then changed the force law slightly to $F \propto 1/r^{2.1}$. The true period exponent becomes 1.55 instead of 1.5 (I confirmed this directly: the slope of $\log T$ vs $\log a$ was 1.5500). I added a power operator to PySrs possible operators ($\wedge$) and PySR immediately recovered

$$T \approx a^{1.55}. \quad (3)$$

6 Can it discover a conserved quantity?

I then tried something harder: instead of fitting a curve, I gave PySR the raw orbital state (x, y, v_x, v_y) from several orbits and asked it to find a quantity that stays constant. For **angular momentum** it succeeded, finding

$$L = x v_y - y v_x \quad (4)$$

exactly. For **energy** ($\frac{1}{2}v^2 - GM/r$) it *failed* — it could not build the $1/r$ term and settled for a poor position-only fit. But when I handed it the right physical variables r and v^2 instead of raw coordinates, it found the energy law easily. So the hard part was not the search, but was choosing the right variables. This matches the idea behind methods like AI-Feynman: symbolic regression only finds structure when the variables make that structure simple.

7 Real data: stellar surface gravity

Finally I applied symbolic regression to real APOGEE survey data (the same data as Problem Set 1). I asked PySR to predict a star's surface gravity (LOGG) from its temperature (TEFF) and metallicity (M.H) for red-giant stars. It found a simple relationship — mostly a linear dependence

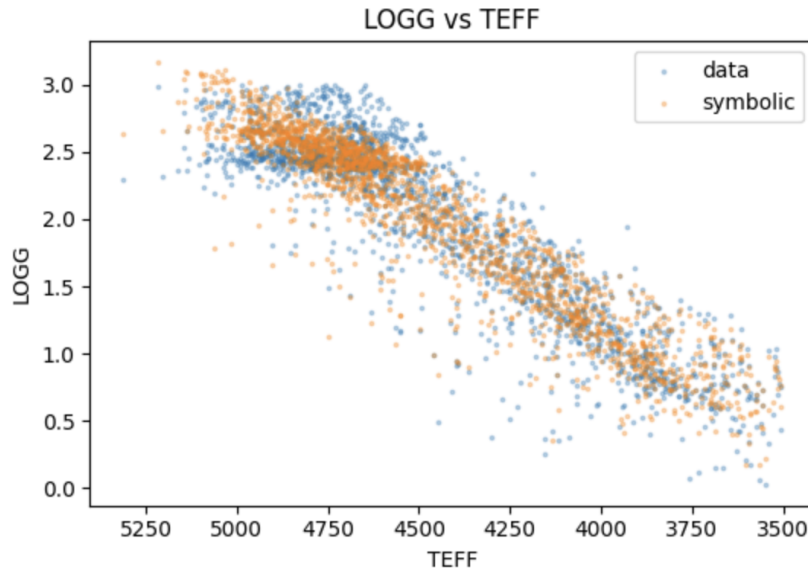


Figure 1: LOGG vs TEFF

on temperature, with a small metallicity correction. The symbolic fit had an error of 0.183 dex,

compared to the raw spread of 0.685 dex in the data, so the equation captures most of the real variation. The leftover scatter is genuine astrophysics (differences in mass and evolutionary stage) that a two-variable law cannot capture. Unlike the Kepler tests, this answer was not known in advance.

8 Conclusion

Symbolic regression reliably rediscovers known laws from clean data, and is fairly robust to noise. Its main limitations are practical: it can only find laws expressible with its operators, and it works far better when given sensible physical variables. On real stellar data it distilled a noisy relationship into a simple, interpretable equation, which is exactly the kind of thing it is useful for.